

I1 extends I2, I3

Program Code:

```
package core_java;

interface I2 {
    void turnOn();
}

interface I3 {
    void setVolume();
}

interface I1 extends I2, I3 {
    void playMusic();
}

class C1 implements I1 {

    public void turnOn() {
        System.out.println("Music system is turned on");
    }

    public void setVolume() {
        System.out.println("Volume set to 10");
    }

    public void playMusic() {
        System.out.println("Playing favourite song");
    }

    void turnOff() {
        System.out.println("Music system is turned off");
    }
}

public class I1toI2I3 {
    public static void main(String[] args) {

        C1 obj = new C1();

        obj.turnOn();
        obj.setVolume();
        obj.playMusic();
        obj.turnOff();
    }
}
```

Explanation:

This program demonstrates multiple interface inheritance. Interface I1 extends both I2 and I3. Therefore, I1 contains the methods inherited from I2 and I3 along with its own playMusic() method. Class C1 implements I1 and provides implementations for all interface methods. It also contains its own turnOff() method.

Compilation and Execution Commands:

Save the file as: I1toI2I3.java

Compile:
javac -d . I1toI2I3.java

Run:
java core_java.I1toI2I3

Output Screenshot:

```
Music system is turned on
```

```
Volume set to 10
```

```
Playing favourite song
```

```
Music system is turned off
```

Verification:

The output confirms that class C1 successfully implements the methods inherited by I1 from both I2 and I3, as well as the method declared directly in I1.

Result:

Thus, the program was successfully executed and the multiple interface inheritance relationship I1 extends I2, I3 was verified successfully.